

MATERIAL HONESTY MEETS MODULAR LONGEVITY:

Designing Gaming Peripherals for Performance,
Identity, and Planetary Limitations

A Speculative Innovation Plan

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DES363: Innovation and Ideation

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Syracuse University, Fall 2025

THE GAMING PERIPHERAL PARADOX

It's 2:00 AM, and Marcus is about to click "checkout" on his third gaming mouse in eighteen months. The current one works perfectly, but Logitech just released a new model with 26,000 DPI and a limited-edition white colorway. He hovers over the purchase button, feeling torn between wanting the latest technology and knowing his current mouse will join the pile of old electronics destined for a landfill where the plastic will take 500+ years to degrade.

Marcus isn't unusual. He's part of a \$4.5 billion global industry built on endless replacement cycles driven more by cultural desire than actual device failure. In gaming culture, your mouse and keyboard aren't just tools. They're extensions of competitive identity, signals of dedication, markers of belonging. But here's the paradox: Gen Z makes up 35% of gaming peripheral purchases and consistently reports sustainability as a core value. 73% say they're willing to pay premium prices for sustainable products (McKinsey & Company, 2023), yet they still upgrade peripherals every 2-3 years on average, generating an estimated 50 million units of annual e-waste.

This isn't hypocrisy. It's a design failure. The gaming peripheral industry doesn't give environmentally conscious gamers any way to reconcile their performance demands with their environmental values. Want cutting-edge technology? Buy a new mouse. Want to reduce e-waste? Accept obsolete performance. Want sustainable materials? Choose budget options with inferior build quality. The current market treats these desires as mutually exclusive.

This project asks: what if they're not? This speculative design intervention addresses Option 4 from the course framework (Emerging Subcultures, Materials, or Behaviors that Suggest a Coming Shift) combined with Option 3 (New Delivery Systems for Health, Food, or Data). Following Option 4's invitation to "act as design anthropologists" and "identify a micro-trend, subculture, or material practice that reveals how people are reimagining value, identity, or belonging," this project looks at gaming peripherals as a space where emerging sustainability behaviors are colliding with performance culture. Drawing from Option 3's focus on "infrastructure as design" and how "packaging, portability, and distribution can be as innovative as the product itself," I'm reimagining the entire delivery system for accessing gaming technology, not through better individual products, but through modular infrastructure designed for longevity.

SECTION I: CONTEXT

The gaming peripheral market is projected to reach \$7.2 billion by 2028 (Grand View Research,

2024), and it thrives on replacement cycles that have nothing to do with product failure. The difference between this year's flagship mouse and last year's is often imperceptible. A 1.5% sensor improvement that human hand precision can't even utilize. Yet gamers keep upgrading anyway, driven by several factors: the perception that equipment is a prerequisite for being serious about gaming, limited edition culture borrowed straight from Supreme's playbook, content creator influence where new releases generate content that drives purchases, and aesthetic identity where gaming setups become carefully curated Instagram content.

Current peripherals embody linear "take-make-dispose" economics. They use virgin ABS plastic, and even when manufacturers incorporate 10-15% recycled content, they hide it in black colorways. They're built with sealed construction using glued components (iFixit consistently scores them 3-4 out of 10 for repairability). They offer zero upgradability, so when sensor tech advances, the entire mouse becomes obsolete. And they provide no end-of-life management, so most just end up in landfills. The result? Devices with 5-7 year functional lifespans getting replaced every 2-3 years.

	TRADITIONAL MOUSE	NEXUS MODULAR SYSTEM
MATERIALS	Virgin ABS plastic Petroleum-derived	85% recycled plastic Visible material story
CONSTRUCTION	Glued, sealed Proprietary parts	Magnetic, modular Standardized parts
REPAIRABILITY	3/10 (iFixit)	9/10 (estimated)
UPGRADE PATH	None, replace entire unit	Core every 3-5 yrs (\$45) Shell anytime (\$29-34)
LIFESPAN	2-3 years	10+ years
END-OF-LIFE	Landfill (500+ yrs)	85% returned, reprocessed
CO ₂ IMPACT	4.8kg per unit	1.6kg (saves 3.2kg)
10-YEAR COST	About \$400 (5 mice)	About \$334 (1 system)

Industry sustainability efforts so far have been pretty superficial. Logitech publishes carbon labels but doesn't offer take-back programs. Razer uses ocean plastic, but only in specific

product lines, and they present the recycled content apologetically instead of celebrating it. SteelSeries focuses on packaging, which represents less than 5% of a peripheral's total lifecycle impact, while the actual products remain unchanged. Meanwhile, Framework Laptop has proven that modular consumer electronics can be both viable and profitable (they score 10/10 on repairability), yet no peripheral manufacturer has tried applying these principles to mice or keyboards.

This creates real tension for Gen Z gamers who value sustainability, authenticity, performance, and uniqueness, but are being forced to choose between disposable high-performance products or inferior sustainable alternatives. This situation parallels themes we've studied throughout the course. From Supreme in Act I, we learned how scarcity creates emotional demand beyond function, but gaming peripherals adopted this model without the ethical constraint. They limit colorways while pushing unlimited replacements. From Disney in Act I, we studied total experience ecosystems, and gaming followed this path with coordinated "battlestation" setups, but Disney's immersion is designed to last while gaming's ecosystem forces constant replacement. From Warstic in Act I, we explored how authenticity and narrative can compete with industrial scale, but gaming peripherals hide their manufacturing story and obscure material origins. And from Act II's systems like MREs and World Central Kitchen, we observed design responding to constraint. Gaming peripherals now need to be designed for material constraint (virgin plastic scarcity, e-waste crisis) without compromising on gaming performance, the same way MREs were designed for extreme constraint without compromising soldier nutrition.

SECTION II: OPPORTUNITY SPACE

Following Option 4's directive to "act as design anthropologists," I've been observing where gaming culture is already starting to shift. These micro-trends represent what the assignment describes as "small behaviors today [that] often signal much larger shifts in culture."

First, there's the mechanical keyboard modding renaissance. In just the past five years, custom mechanical keyboards have gone from a niche hobby to mainstream gaming culture. Communities like r/MechanicalKeyboards have over 600,000 members, and these are people spending \$300-800 on keyboards they assemble themselves. They're willing to accept higher upfront costs because of longevity (hot-swap switches mean never buying a new board), imperfect aesthetics (you can see screws and exposed components), complexity (some build guides are 40-minute YouTube tutorials), and they actually prefer uniqueness over uniformity because no two builds end up identical. Group buys for custom keycap sets regularly raise \$500K to \$1M, with wait times of 6-12 months. This isn't impulsive consumption. It's people investing in objects designed to last. This micro-trend shows how gamers are "reimagining value" (to use Option 4's language) by prioritizing longevity and personalization over novelty.

Second, there's the solarpunk aesthetic movement. This is emerging primarily among Gen Z on TikTok and Instagram, and it represents a crucial shift from climate doom to climate action. Instead of "we're destroying the planet," it's "we're building a future where technology and nature coexist." The aesthetic markers are visible organic materials, technology that looks alive rather than sterile, warm earthy tones replacing the typical RGB and all-black gamer aesthetic, and transparent materials that show honest construction. Solarpunk gamers are modding

their setups with bamboo wrist rests, 3D-printing planters for their monitors, and choosing keyboards with wooden cases. They're actively seeking products where sustainability is visible and tactile, part of their aesthetic identity rather than hidden behind greenwashing marketing. The hashtag #solarpunk has over 500M TikTok views, with gaming setup tours prominently featuring sustainable materials. Users specifically call out products that "hide" sustainability versus those that "celebrate" it. This shows how people are "reimagining identity" (Option 4 again) where sustainability becomes an aesthetic statement, not a sacrifice.

Third, material transparency has become a trust signal. Partly in response to greenwashing scandals, Gen Z increasingly treats material transparency as a baseline expectation. They don't just want products labeled "eco-friendly." They want to know the exact materials (percentages, sources), where those materials came from (specific suppliers, regions), and what happens at end-of-life (actual recycling pathways, not just theoretical ones). You can see this in fashion apps like Good On You that score brands on transparency, food products with blockchain-tracked supply chains gaining premium pricing, the Right to Repair movement demanding access to schematics, and "material passport" concepts gaining traction in EU policy. People are literally googling "what is this made from" before purchasing electronics. They're learning terms like "virgin plastic," "post-consumer recycled," "biodegradable vs. compostable." This isn't niche environmental activism anymore. It's mainstream consumer literacy. This micro-trend demonstrates how people are "reimagining belonging" (Option 4) where community forms around shared values of transparency and accountability.

These three trends converge on the same cultural shift that Option 4 describes: "Every generation creates its own design language, often shaped by the materials it touches, the tools it uses, and the values it inherits or resists." Gaming culture is clearly ready to reward authenticity, transparency, and longevity if products are designed with these values from the start rather than trying to retrofit them apologetically. The challenge, as the assignment puts it, is "to translate cultural observation into design speculation, to imagine what these emerging practices might become tomorrow."

Looking at where current design prevents the behaviors these communities are trying to enact (what the assignment calls places where "people's needs, contexts, and systems misalign"), I've

identified four key tensions. First, upgrade culture versus environmental impact. Gamers want cutting-edge specs, manufacturers release yearly updates, and entire products become obsolete. The opportunity here is separating upgradeable performance components (sensors, switches) from durable structure and customizable aesthetics. The keyboard community already does this, so why not extend it to all peripherals?

Second, sustainability feels cheap or invisible. Recycled plastics are either hidden in black colorways (trying to look "new") or used in budget products, which reinforces the perception that recycled equals lower quality. The opportunity is reframing recycled materials as premium, unique, and authentic. We can position the visible signs of recycled content as desirable markers of authenticity, connecting to wabi-sabi philosophy and Warstic's craft narrative.

Third, black box design hides material origins. Users have no idea what their peripherals are made from, where materials came from, or what happens at end-of-life. The opportunity is making material origins visible and trackable. Each shell's unique recycled content pattern becomes its "fingerprint," and digital material passports provide transparency without overwhelming users.

Fourth, planned obsolescence versus emotional attachment. Products are designed to be replaced, not cherished. Clean, uniform aesthetics work against patina because scratches just look like damage. The opportunity is designing products that actually improve with use. Textured recycled materials hide fingerprints, surface variation makes scratches less visible, serial numbers and unique patterns create collectibility, and modularity means users can keep their favorite shell while upgrading internals. This builds attachment instead of encouraging replacement.

SECTION III: DESIGN INTERVENTION

The Nexus System is a modular gaming peripheral platform where the architecture separates upgradeable performance from durable structure and customizable aesthetics. Recycled materials get celebrated through visible texture and unique patterning. Material transparency becomes part of product identity through digital passports. And users build collections of shells rather than replacing entire devices.

This addresses Option 3's focus on "infrastructure as design" and "rethinking systems of care and connection." Rather than just designing better individual mice, I'm reimagining the entire delivery system for accessing gaming technology. Like the assignment's examples where "packaging, portability, and distribution can be as innovative as the product itself," the real innovation here is in how the system delivers performance over time through modular infrastructure rather than through disposable units.

The core design principle is pretty simple: instead of fighting against the visual reality of recycled materials, embrace it. "Your mouse carries the signature of its source. Ocean plastic's teal speckles, e-waste's metallic flecks, bioplastic's organic texture. No two shells are identical. This isn't imperfection. It's authenticity."

This approach takes what Option 1 calls "digital honesty" and applies it as material honesty, where the imperfections of recycled content become forms of value rather than flaws to hide. Drawing from the assignment's reference to "Jack White's obsession with imperfection" and "Warstic's narrative of craft as identity," the goal isn't nostalgia for pre-digital manufacturing. It's a forward-looking design philosophy where visible material history signals authenticity and trust. The assignment asks: "how might designers reintroduce truth, tactility, or trace into digital experiences?" Here's how: truth through material passports showing origin, tactility through textured recycled surfaces you can actually feel, and trace through visible signs of the material's previous life.

The product uses a three-layer architecture. The core module contains all the upgradeable electronics: a PixArt sensor with 26K DPI, Omron switches rated for 100 million clicks, wireless plus wired connectivity, and 80-hour battery life. You upgrade this every 3-5 years for \$45 (with a \$15 trade-in credit). The performance matches current flagship mice.

The frame is the permanent piece made from recycled aluminum. It weighs 68 grams, uses magnetic shell mounting that's completely tool-free, and is designed to last 10+ years. It's included in your initial purchase.

The shell is the swappable aesthetic layer. There are three material series (I'll detail these in a sec), they attach magnetically and you can swap them in 30 seconds, each one has a unique serial number, and they cost \$29-34 each.

The initial bundle is \$89, which gets you the frame, core, and your choice of shell. Additional shells are \$29-34 each. Core upgrades are \$45 every 3-5 years with that \$15 trade-in. The system also extends to mechanical keyboards using identical principles: hot-swappable switches, recycled aluminum frame, same three material series for keycap sets. Keyboard pricing is \$119 for the core unit and \$59-69 for keycap sets.

The three material series follow the assignment's observation that "every generation creates its own design language, often shaped by the materials it touches." These represent Gen Z's emerging material practice, one that celebrates rather than hides recycled content.

The Ocean Series costs \$29 and uses 85% post-consumer ocean plastic plus 15% bio-binders. Visually, you get a deep blue base with teal, white, and sand speckles, and literally no two are identical. Pattern variance ranges from teal-heavy versions (about 15% of production, these become collector favorites) to balanced versions (65%, most common) to sand-heavy versions (20%, coastal aesthetic). Each shell prevents 0.5kg of ocean pollution and reduces CO₂ emissions by 3.2kg compared to virgin plastic. The material passport shows you the specific cleanup source, like "Thai coastal initiative, November 2025." This appeals to environmentally conscious gamers and people who like coastal aesthetics. Serial numbers look like OS-2847.

The Circuit Series also costs \$29 and uses 90% recycled e-waste plastics plus 10% recovered metallic particles like copper, aluminum, and trace amounts of gold. Visually, it's charcoal gray to black with metallic flecks that catch the light. You get silver, copper, and gold particles visible throughout. Pattern variance includes silver-dominant versions (50%), copper-heavy versions (30%, these are rarer with a warm tone), and circuit-visible versions (20%, where you can actually see PCB patterns, which tech enthusiasts love). Each shell repurposes materials from about 3 obsolete devices in a true closed-loop where gaming tech becomes gaming tech again. The material passport shows the source type, like "Desktop PC plastics, Boston e-waste collection." This appeals to people into cyberpunk aesthetics, tech enthusiasts, and competitive gamers who want that industrial look. Serial numbers look like CS-8392.

The Harvest Series costs \$34 (slightly more because bioplastic processing is more expensive) and uses 75% hemp-based bioplastic plus 25% PLA from sugarcane or corn. Visually, you get warm earth tones like sand, wheat, tan, and light brown, with visible plant fiber throughout the material. Pattern variance includes light fiber-visible versions (30%, minimalist look), wheat-tone versions (50%, most popular), and dark fiber versions (20%, rustic handcrafted appearance). The environmental impact is actually net-negative carbon because the plants absorbed CO₂ while growing. It's also biodegradable in industrial composting facilities (6-12 months versus 500+ years for traditional plastic). The material passport shows the crop source, like "Hemp fiber from Oregon agricultural cooperative." This appeals to people into solarpunk aesthetics, minimalist setups, and nature-positive gaming. Serial numbers look like HS-4521. Unlike the other series, at end-of-life you can either return it for recycling or send it to industrial composting.

The design philosophy across all three series is that the speckles, flecks, and fiber textures aren't flaws. They're signatures that prove where the material came from. These "imperfections" serve multiple functions: they're proof of the material's previous life, they're uniqueness markers (OS-2847 literally looks different from OS-2848), they drive collectibility through natural material variation instead of artificial scarcity, they spark conversations (visible recycled content prompts questions), and they age gracefully because textured surfaces hide

minor scratches way better than smooth virgin plastic.

Every shell includes a material passport system. There's a physical component: a laser-etched serial number (like "OS-2847") and a QR code on the underside that links to a digital passport. The digital passport is a mobile app with three core screens.

Screen one is registration and overview. It shows a high-res photo of YOUR specific shell's unique pattern that was captured during manufacturing. You see the serial number, registration date, and current owner. There's a material breakdown with progress bars showing things like "85% post-consumer ocean plastic, 15% bio-binders." Source documentation tells you exactly where it came from, like "Thai coastal cleanup, August 2026." And environmental impact metrics show you the CO₂ saved versus virgin plastic, ocean plastic prevented, and the overall return rate for recycling.

Screen two is history and upgrades. It's a vertical timeline showing shell registration, core upgrades with dates and specs, and frame milestones. Each entry includes transaction details and your cumulative environmental impact. Available upgrades show up here with clear pricing (the trade-in credit is already applied), and you can compare your current components to new ones.

Screen three is community and trade-in. There's a photo gallery where users share their unique patterns and you can discover "twin" shells that have similar color distributions. Collection management shows all the shells you own with their individual trade-in values. The trade-in process is streamlined with prepaid return labels that generate automatically. And there's messaging like "OS-2847 will become part of 2032 production shells" to emphasize the circular nature.

This passport system represents what Option 3 calls "infrastructure as design." It's not the product itself, but the system for tracking, upgrading, and returning materials that actually enables circularity. Like the assignment's examples where "distribution can be as innovative as the product itself," the real innovation in the material passport is how it delivers transparency and facilitates material flows.

The user experience works like this. You purchase either the full bundle (core plus frame plus shell for \$89) or you buy components separately. You choose your material series based on aesthetic preference and values, and you receive the product with its unique serial number. For unboxing, following Option 3's principle that "packaging, portability, and distribution can be as innovative as the product itself," the packaging actually transforms into functional desk infrastructure. The mouse box becomes a desk organizer and the keyboard box becomes a

cable management tray. Everything uses mushroom-based packaging or compressed recycled paper. No plastic wrap. The setup card shows you how to attach the shell and links you to the digital passport.

Daily use is straightforward. The textured recycled surface provides better grip than smooth plastic. The matte finish resists fingerprints. The unique pattern makes your device instantly recognizable on your desk. And the performance matches or exceeds current flagship mice. When you want to customize, you buy additional shells in different materials for \$29-34 each. You can swap them tool-free in 30 seconds to match your desk aesthetic, the game you're playing, or just your mood. You build up a collection over time.

When it's time to upgrade (when new sensor technology releases every 3-5 years), you buy the core module upgrade for \$45 with the \$15 trade-in, which nets out to \$30. You keep your existing frame and your entire shell collection. Installation takes about 2 minutes with a video guide. At true end-of-life (when you're ready to move on from the system entirely), you return shells via mail using a prepaid label. You receive credit toward new shells or a complete new system. The shells get reprocessed into the next generation of products. The frame is recyclable through municipal aluminum recycling. And the core has a proper e-waste recycling pathway.

The business model makes sense financially. Revenue comes from initial hardware (\$89 bundles, which have higher margins than typical peripherals), ongoing shell sales (\$29-34, creating continuous revenue as users build collections), core upgrades (\$45 every 3-5 years), and keycap sets (\$59-69, where the collectible nature drives multiple purchases).

Comparing profit models: the traditional approach is selling an \$80 mouse every 2-3 years, which works out to \$27-40 per year per customer. The Nexus model is \$89 initial plus a \$30 shell every 8 months plus a \$45 upgrade every 4 years, which works out to \$45+ per year per customer. Plus, you have way lower customer acquisition costs because retention replaces the constant need for new customers.

Why this works financially: the higher initial price is justified by longevity plus the uniqueness factor. Shell collecting creates ongoing revenue instead of one-time purchases. Modular manufacturing is actually cheaper because you have standardized cores and simple shell molds. Material costs work in your favor since recycled plastics are often cheaper than virgin plastics. And customer lifetime value is much higher with 10+ year relationships versus 2-3 year relationships.

This intervention synthesizes lessons from Acts I and II, which the assignment framework requires. From Supreme in Act I, it creates desire through uniqueness, but it's authentic uniqueness (actual material variation) rather than artificial scarcity (manufactured production

limits). As we studied, Supreme's "economics of scarcity and cultural capital" create "emotional demand far beyond function," and here that's being applied through genuine material variation instead of just limiting production.

From Disney in Act I, it builds a total experience ecosystem, but one designed for longevity instead of obsolescence. Where Disney "expanded into total experiences, transforming consumption into immersion," Nexus creates immersion through accumulated shells and upgrades over years rather than through replacement cycles.

From Warstic in Act I, it embodies the craft philosophy with visible origins, celebrated "imperfection," and story embedded in the material itself. As the assignment notes, this demonstrates "how authenticity and narrative can compete with industrial scale, how craft and story become an alternative form of innovation."

From MREs and emergency systems in Act II, it responds to constraint (virgin plastic scarcity, e-waste crisis) without compromising performance. It's engineering for limitation while maintaining function. As Act II revealed, design becomes "about function under pressure, how form, delivery, and technology intersect when the goal is to nourish, protect, or stabilize communities." Here that's being applied to stabilizing gaming communities within environmental constraints.

The synthesis is this: sustainable design isn't about sacrifice. It's about applying Act I's meaning-making principles (scarcity, experience, authenticity) within Act II's material constraints, creating what Option 3 calls "infrastructure as design" that rethinks "systems of care and connection." Human desires for performance, aesthetics, and identity become pathways toward sustainability rather than obstacles to it.

SECTION IV: IMPLEMENTATION AND FUTURES

Making this real would require some changes, but nothing impossible. For material sourcing, the partners already exist. Organizations like Ocean Conservancy and 4Ocean are already collecting ocean plastic, they just need processing partnerships. Electronics recyclers are already shredding and separating plastics, they just lack buyers. Agricultural waste processors are already handling hemp, corn, and sugarcane for other industries. The technical requirements all use existing technology: injection molding works fine with recycled content (it's a standard process), magnetic mounting is already used in phone cases, hot-swap PCB sockets are already standard in enthusiast keyboards, and digital passport infrastructure is just QR codes plus a database, which is relatively simple. The challenge here is integration, not invention. Nothing actually needs to be invented.

There are cultural barriers, but each one has a clear counter-argument. "Gamers won't accept eco products" gets countered by the fact that gamers clearly accept uniqueness, as we see with custom keycaps. The solution is leading with performance and uniqueness where sustainability is a bonus, not the main selling point. "Recycled looks cheap" gets countered by pointing out that current products only hide recycled content because they're trying to make it look "normal." The solution is designing from scratch for the recycled aesthetic, like how raw denim or wabi-sabi ceramics work. "Modular means complicated" only happens if it's poorly designed. The solution is tool-free magnetic assembly with 30-second swaps, and Framework has already proven that modular can be simple. "Can't compete with major brands" assumes you need to compete on volume, but you don't. The solution is competing on value proposition, not volume, using a direct-to-consumer model like Glorious Gaming.

The regulatory environment actually favors this approach. The EU Right to Repair directive from 2023 onward requires spare parts availability. France's repairability index mandates scoring. Extended Producer Responsibility legislation makes manufacturers financially responsible for end-of-life. US states are passing similar laws (Repair.org, 2024). Modular systems position you ahead of these regulations rather than just reacting to them.

The go-to-market strategy has three phases. Phase 1 in Year 1 is an enthusiast community launch. Partner with gaming content creators, focus on the mechanical keyboard community first since they already value modularity, use Kickstarter or crowdfunding for initial production with a goal of raising \$500K to \$1M, and start with just 10,000 units in limited material series (Ocean plus Circuit). Phase 2 in Year 2 is material expansion. Introduce the Harvest series, add the mouse product line, launch the material passport app with community features, and begin forming retail partnerships. Phase 3 in Year 3 and beyond is the mainstream push with broader SKU availability, lower-cost entry products, major retailer partnerships, and corporate responsibility partnerships.

Looking at the speculative timeline, from 2026 to 2030 is the early adopter phase. The Kickstarter launch raises \$500K to \$1M from the enthusiast community. The first 10,000 units ship direct to consumers. The mechanical keyboard community leads adoption. Gaming YouTubers start reviewing repairability scores alongside performance specs. Shell-swapping videos trend on TikTok. The first "shell trading" communities form online. The key indicator of success would be at least one major manufacturer offering modular components by 2030.

From 2030 to 2035 is mainstream adoption. Modular peripherals reach 25-30% of the enthusiast market. Price parity gets achieved where modular costs the same as traditional when you factor in total cost of ownership. Major retailers dedicate shelf space to "Sustainable Gaming" sections. The shell secondary market reaches \$50M+ annually. Culturally, the "virgin plastic aesthetic"

becomes increasingly outdated. Material passports become an expected feature, not a differentiator. Esports teams start adopting modular systems as part of sustainability partnerships. Regulatory-wise, the EU mandates material passports around 2032, and US states start requiring minimum recycled content percentages. The key indicator would be modular architecture becoming the default expectation where reviewers actively criticize sealed designs.

From 2035 to 2040 is when it becomes the system standard. 60-70% of peripherals sold use upgradeable cores. Standardized mounting systems emerge, kind of like how USB standards work. The shell return rate hits 85%+ and achieves true closed-loop maturity. Culturally, Gen Alpha only knows modular systems because that's all they've ever used. Material provenance becomes a status symbol. You see cross-generational usage where parents pass frames down to their children. The key indicator would be that disposing of a working peripheral becomes culturally unthinkable, the same way we now view throwing away a quality watch or chef's knife.

This could spark a broader Material Honesty Movement. Success in gaming peripherals catalyzes application to smartphones (with modular components), appliances (with 30-year frames and upgradeable controls), fashion (with visible recycled textiles), and furniture (with component tracking through multiple lifecycles). By 2040, the practice of celebrating rather than hiding recycled content becomes a core design principle taught right alongside form and function. As the assignment asks us to imagine: "what these emerging practices might become tomorrow." This represents that future, where the micro-trends that I identified earlier (keyboard modding, solarpunk aesthetics, material transparency demands) have scaled from subcultures to design standards.

CONCLUSION

Gaming peripherals seem like an unlikely place to try redesigning consumption itself. They're not life-or-death like medical devices. They're not infrastructure-critical like transportation. They're literally just the tools we use to play games. But that's actually precisely why they matter.

If we can only design sustainable systems for the "important" objects, the ones where people will tolerate sacrifice, then we're never going to achieve the kind of systemic change that the climate crisis requires. The hard challenge isn't the 10% of products where sustainability can be mandated or legislated. It's the 90% of objects that are bound up with identity, joy, and community.

Gaming peripherals test that proposition. They're technology products subject to rapid advancement. They're cultural artifacts that signal belonging. They're aesthetic objects that get carefully curated and displayed. They're performance tools where compromise feels completely unacceptable. If modular architecture, recycled materials, and material transparency can work here, in one of consumption's most psychologically complex categories, then they can probably work almost anywhere.

Looking back at Acts I and II helps clarify what Act III is attempting. Act I showed us how design creates desire through scarcity, immersion, and authenticity. Those examples revealed that emotional value often matters way more than functional value. The story of how something is made can actually outweigh its utilitarian purpose. Act II shifted our focus from desire to necessity, from consumption to survival systems. Through examples like MREs and emergency food systems, we saw design responding to constraint. We learned how limitation can actually focus design rather than prevent good design.

Act III asks the synthesis question: what happens when we apply Act I's lessons about meaning-making to Act II's reality of constrained resources? What if we treat environmental constraint not as a sacrifice we need to minimize, but as a creative boundary to work within?

The Nexus System tries to attempt that synthesis. It takes Supreme's scarcity economics but applies them honestly where shells are unique because materials genuinely vary, not because production is artificially limited. It uses Disney's total experience approach but designs it for longevity instead of obsolescence. It embraces Warstic's craft philosophy completely with visible origins and celebrated "imperfections." And it responds to constraint the way MREs did, addressing virgin plastic scarcity without compromising gaming performance.

The result is this: sustainable design isn't actually about sacrifice. It's about synthesis. We don't need users to care less about performance or aesthetics. We need to build systems where those very human desires lead toward sustainability rather than away from it.

Three factors suggest this could actually be viable. First, cultural readiness. The micro-trends I identified earlier (mechanical keyboard modding with 600,000+ members paying premium prices for modularity, solarpunk aesthetics with 500M+ TikTok views, material transparency demands where Gen Z treats transparency as a baseline expectation) prove that gaming culture will reward authenticity, transparency, and longevity if products are designed with these values from the start. These represent what the assignment describes as "small behaviors today [that] often signal much larger shifts in culture."

Second, regulatory momentum. The EU Right to Repair directive, France's repairability index, Extended Producer Responsibility legislation, and state laws being passed in California and New

York all create an environment where manufacturers who wait to comply reactively are going to face a competitive disadvantage against those who design proactively.

Third, technical feasibility. Literally nothing in this system requires invention. Modular assembly, magnetic connections, recycled material processing, material passports, take-back logistics. All of these exist and function in adjacent industries already. Framework has proven that laptops can be fully modular and commercially successful. The challenge isn't technical innovation. It's business model courage.

This brings up the bigger question: can sustainability actually become identity? This project started by asking whether gaming peripherals can be sustainable without sacrificing performance or identity. But I think the right question is actually flipped: can sustainability become identity?

Can the recycled speckles in your mouse shell become as desirable as a Supreme box logo? Can they become markers that you understand value beyond just novelty, that you participate in systems designed for longevity, that your aesthetic choices actually align with your stated values? Can knowing your keyboard's material origin story become as satisfying as knowing its switch specifications? Like knowing where the hemp was grown, which cleanup initiative your ocean plastic came from, what circuit boards your metallic flecks originated from? Can keeping the same frame for ten years become a point of pride rather than embarrassment? Can it prove that you invest in quality, that you maintain relationships with objects, that you resist the replacement culture treating everything as temporary?

Based on the micro-trends already emerging in gaming culture, I think the answer is starting to form: yes. Not someday in some hypothetical future. Now. Today. In communities that are already trying to enact these values despite market structures actively working against them.

What's missing isn't cultural readiness. It's products designed from the ground up to make those values actually actionable. It's the "infrastructure as design" that Option 3 talks about, the kind that enables these behaviors to scale beyond just small communities.

This speculative design intervention proposes gaming peripherals as a beginning, not an ending. If gamers end up embracing shells with recycled material signatures, if modularity becomes an expected feature rather than an exception, if material honesty becomes the new premium standard, then we'll have proven something important. We'll have proven that sustainability doesn't require sacrifice.

We'll have shown that the future isn't necessarily about less consumption (though that might be necessary too). It's about different consumption. Consumption that's designed for longevity. Consumption where the story actually matters. Consumption that creates relationships rather than

just transactions.

We'll have demonstrated that Gen Z's supposed paradox (wanting both performance and sustainability, belonging and responsibility) isn't actually a paradox at all. It's a design challenge. And it becomes totally solvable when we stop treating competing values as mutually exclusive and start treating them as creative constraints that focus our thinking rather than limit it.

The gaming mouse with its speckled ocean plastic shell, its modular core, and its decade-lasting frame isn't just a product. It's a microcosm of what sustainable technology could actually be. It's fundamentally different in its relationship to materials, to users, and to time itself.

It's proof that design innovation in 2026 doesn't mean inventing new technologies. It means redesigning the stories, systems, and structures that either turn human nature into an obstacle or into a solution. As Option 4 asks us to do: "translate cultural observation into design speculation, to imagine what these emerging practices might become tomorrow."

Gaming peripherals seem pretty trivial when measured against the planetary crisis we're facing. But they're actually not trivial at all. They're the test case that could prove sustainable design can be desirable design. They're the beginning of material honesty becoming material value. They're how we might learn to design for abundance within limits.

And honestly, that might be the most important design skill our generation needs to develop.

References

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This project was developed with assistance from Claude, an AI assistant created by Anthropic. Claude contributed to research organization, writing support, product system specifications, and conceptual development. All final decisions regarding project direction, design principles, and framework were made by the author.

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